

Vantage VTG-05: A Rapidly Deployable Magnetic Pole-Climbing Robot for Elevated Surveillance

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Abstract: Ground commanders responding to a terror or public-order incident want situation awareness in the first hour, yet the last mile of surveillance deployment is still manual. Officers strap a camera and its hoist onto a pole by hand, in the manner of existing hoisted camera systems, so set-up is slow and trained personnel are required at exactly the moment when manpower is stretched thin. This paper presents VTG-05, a magnetic-wheeled climbing robot that carries a Singapore Police Force body-worn camera up a ferromagnetic lamppost and provides a remotely commanded elevated view. The vehicle runs on two N52 magnet-array wheels. Each wheel is driven by a pair of N20 gearmotors for double torque and steered by its own MG996R servo, so heading on the pole is corrected by yawing the wheels rather than by differential speed. An 8x8 time-of-flight ranging array provides closed-loop pole alignment, an inertial measurement unit supervises chassis attitude, and an ESP32 coordinates both over ESP-NOW. A magnetic-adhesion model sized the wheels: for a 1.5 kg target mass, a rubber-on-steel friction coefficient of 0.5 and a safety factor of 1.5, a total magnetic pull force of 44.15 N is required, which is met by N52 arrays equivalent to a 5.21 mm square pole face per wheel and delivering 2.249 kgf. Testing against the Singapore Police Force operational requirements returned a measured payload capacity of 1.27 kg against a 500 g target, a 4 m climb in 48 s against a 5 min target. Worst-case load capacity across the three grip conditions was 1.27 kg, a safety margin of 2.5 times, and the greatest height demonstrated was 5.77 m. All seven engineering requirements were met, and the platform has been trialled with two industry partners across three demonstration days.

Index Terms: climbing robot, magnetic adhesion, pan-tilt payload, public safety, situation awareness, time-of-flight sensing.

I. INTRODUCTION

The hour immediately after a terror attack or major public-order incident is when a ground commander has the fewest facts and the most decisions to make, judging cordon geometry, casualty triage and crowd movement from fragmentary radio reports; the fastest route to situation awareness is a view from above the crowd. That view is technically easy to obtain and operationally hard to deploy, since the last mile of surveillance is still manual: Singapore streets are already lined with ferromagnetic lampposts and the Singapore Police Force (SPF) already issues body-worn cameras (BWCs) that stream usable video, yet officers still strap the camera and its hoist to a post by hand and run cabling, in the manner of hoisted systems already fielded by the Singapore Armed Forces such as 15C4I (Fig. 1), occupying

two or more trained officers exactly when manpower is stretched thin.

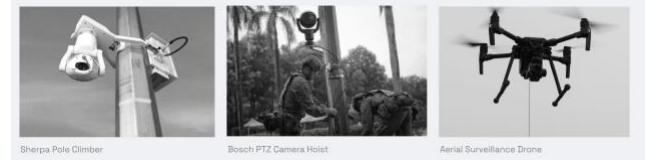


Fig. 1. State of the art: slow set-up and trained personnel required. Left to right, a Sherpa pole climber, manual strapping of a Bosch pan-tilt-zoom camera hoist onto a post, and an aerial surveillance drone.

Three families of prior art were benchmarked, Fig. 1: commercial pole climbers (e.g. the Sherpa) are sized for utility poles and still need an operator at the base throughout set-up; camera hoists (e.g. the Bosch PTZ unit) give excellent imagery once mounted, but mounting is the manual step being attacked; and aerial drones give an unmatched viewpoint at the cost of endurance, a dedicated pilot and airspace clearance. All three share the same weakness: slow set-up and trained personnel required. In the research literature, magnetically adhering climbing robots are established for inspecting large ferromagnetic structures such as ship hulls and steel towers [1]–[3], establishing the two design levers used here, magnetic normal force to weight and wheel-surface friction, but oriented toward flat or gently curved surfaces rather than an operational camera payload on a narrow street lamppost.

The gap is therefore specific. No compact, single-operator, tool-free device takes a 500 g camera from the ground to 4 m on an existing lamppost in under five minutes and then holds station for a full shift. This paper hypothesises that permanent-magnet wheel adhesion, combined with a closed-loop time-of-flight (ToF) steering law that keeps the vehicle centred on the pole, is sufficient to close that gap within the project budget. The three core innovations claimed are the steering mechanism, the pan-tilt mechanism and the point-tilt-zoom payload.

II. PROPOSED APPROACH

A. System Requirements

Requirements were derived from field engagement rather than assumption, through a sequence of engagements with SPF and the Army detailed in Section VI. A Changi Neighbourhood Police Centre visit identified slow deployment and vendor dependence as the core weakness of systems in service, and follow-up meetings and pilot trials refined the design further. Three must-have parameters come directly from the SPF operational requirements: climb 4 m, deploy in under 5 min, and carry a 500 g camera payload, the 5 min figure a 100% improvement on current practice.

Two field measurements fixed the mechanical envelope. The identified octagonal lamppost has a nominal outer diameter of 150 mm, and measurement of a standard Land Transport Authority lamppost gave 70 mm across a flat at 1.5 m height. The smallest square pole in the operating set measures 50 mm across a face. The payload was fixed as the

issued SPF BWC, so that the robot only has to carry and point a camera the officers already use. Table I lists the resulting seven engineering requirements, each written against an objective, measurable parameter and paired with the physical test that verifies it.

TABLE I. REQUIREMENTS TRACEABILITY MATRIX

#	Engineering requirement	Target	Test method
R1	The robot shall carry a payload of 0.5 kg	≥ 500 g	Test maximum load with a luggage scale
R2	The robot should climb with a reliability above 90% for a 2 m vertical climb	$\geq 90\%$	Run the robot up and down ten times with at most one fall
R3	The robot should climb a 4 m pole within 5 min	≤ 5 min	Run from the base to 4 m for three trials and record the time for each successful trial
R4	The unit must climb the identified octagonal lamppost of 150 mm outer diameter without additional tools	150 mm	Climbing test on different octagonal poles
R5	The unit shall be controllable via wireless link up to 10 m	10 m	Move progressively further from the robot until link loss
R6	The onboard power system shall sustain continuous camera and pan-tilt operation on a single charge	≥ 24 h	Run continuous camera and pan-tilt unit operation until battery depletion
R7	The robot shall pan and tilt within a 45 degree cone	$\geq 45^\circ$	Measure the angular displacement of the camera between its pan-tilt extremes

B. Concept Selection

Concept selection was run at two levels: the elevation principle, and then the adhesion and sensing choices within the selected principle.

At the top level the three benchmarked systems in Fig. 1 were compared against the three must-have parameters in Table II. The pole climber and camera hoist satisfy payload and endurance but fail on set-up time and personnel, the costs the project set out to remove. The aerial drone satisfies speed of elevation but fails on endurance, covertness and the need for a dedicated operator. A magnetic climbing robot was selected: the target lampposts are steel, rolling contact is fast, and permanent magnets hold the vehicle without power, so a power loss cannot drop the payload.

TABLE II. COMPARISON OF ELEVATION CONCEPTS AGAINST THE OPERATIONAL MUST-HAVES

Criterion	Sherpa pole climber	PTZ camera hoist	Aerial drone	Magnetic climber
Set-up under 5 min	No	No	Yes	Yes
No trained operator	No	No	No	Yes
Carries 500 g camera	Yes	Yes	Yes	Yes
Endurance on station	Yes	Yes	No	Yes
Holds without power	Yes	Yes	No	Yes
Low acoustic signature	Yes	Yes	No	Yes
Uses existing street furniture	Yes	Yes	n/a	Yes
Requires ferromagnetic pole	No	No	No	Yes

Three design choices within the magnetic-climber concept were worked out through testing across five prototype versions, V1 to V5. For adhesion, the team started with a

pressure-sensitive adhesive tread in V1 and V2, then settled on an N52 permanent-magnet array. For the wheel layout, the design began as a four-wheel platform in V3 and converged in V5 on two larger, independently steered magnetic wheels; this increases the magnetic force at each contact point and avoids the flat-to-flat conflicts that four small wheels create on a polygonal pole. For steering, the team first relied on an inertial measurement unit (IMU) alone, which failed for the reasons given in Section III, and then switched to a time-of-flight (ToF) sensor as the primary input with the IMU as backup.

A fourth decision concerned the payload interface. Rather than integrating a bespoke camera module, the chassis carries the issued SPF BWC in a universal mount, so the robot only has to point it. Because that camera has no optical zoom, the third core innovation substitutes an infrared laser designator and a blob tracker for the conventional pan-tilt-zoom, giving what the team calls point-tilt-zoom.

C. Concept Description

VTG-05 is a two-wheeled magnetic climber with a pan-tilt payload head, shown deployed in Fig. 2. The vehicle is organised into four subsystems: locomotion and steering, the pan-tilt payload, electronics and power, and sensing and control.



Fig. 2. VTG-05 on a lamppost. Left: industrial-design render of the final enclosure with the body-worn camera in its cradle. Right: the functioning prototype during a laboratory climb.

The locomotion subsystem (Fig. 3(b)) uses two 50 mm-diameter, 27.5 mm-wide magnetic wheels, each with an N52 neodymium magnet array behind a rubberised, grip-taped tread. Each wheel is driven by two N20 gear motors, one on each side, coupled through brass couplings, doubling the torque at the contact patch so a two-wheel vehicle can lift the full payload. Each wheel also carries its own MG996R steering servo, correcting heading by yawing the wheels independently rather than by differential speed, which prevents the chassis walking off a curved surface.

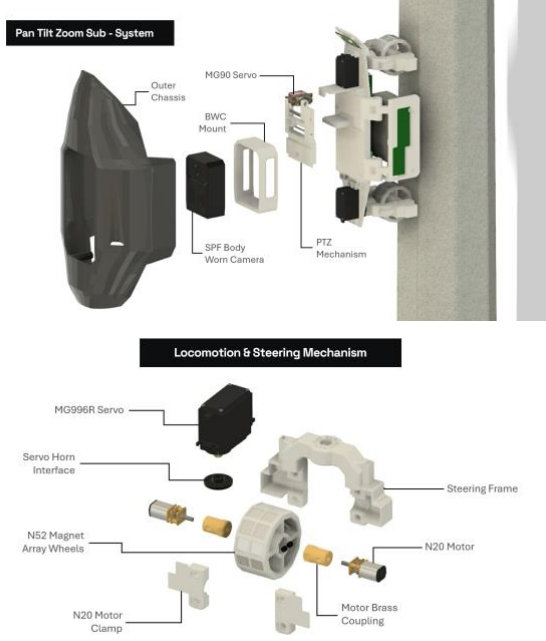


Fig. 3. Exploded assemblies. (a) Pan-tilt sub-system: outer chassis, SPF body-worn camera and mount, MG90 servo and tilt control mechanism. (b) Locomotion and steering: MG996R servo, servo-horn interface, steering frame, N52 magnet-array wheel, N20 motors, brass couplings and motor clamps.

The pan-tilt head, Fig. 3(a), carries the BWC in a printed mount inside the outer chassis. Tilt is driven by an MG90 servo through a tilt control mechanism. Pan is achieved by rotating the whole vehicle about the pole under slow motor control, which is why the braking behaviour analysed in Section III is a payload-pointing problem as much as a safety problem.

Electronics and power sit between the two wheels (Fig. 4(a)) to keep mass close to the pole and the overturning moment small, with the pack built from rechargeable lithium-ion cells in a printed housing with a removable cap. The PCB stack (Fig. 4(b)) comprises an ESP32 with an ultra-wideband module, an N20/servo motor controller built around DRV8833 drivers, a charging circuit and a voltage regulator; sensing comprises a forward-looking ToF array, an IMU, a barometer and a power sensor. Command and telemetry run over ESP-NOW to a browser dashboard offering hold-to-drive forward with automatic pole alignment, hold-to-drive backward with ToF auto-steering, an active-hold brake and an all-motor stop.

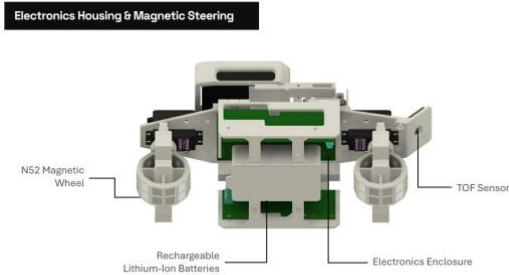


Fig. 4. (a) Electronics housing and magnetic steering, showing the two N52 magnetic wheels with their steering servos, the rechargeable lithium-ion batteries, the electronics enclosure and the forward ToF sensor.

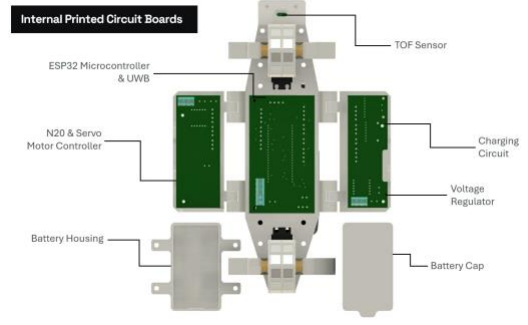


Fig. 4. (b) Internal printed circuit boards: ESP32 and ultra-wideband module, N20 and servo motor controller, charging circuit and voltage regulator.

III. MODELLING

Two modelling activities were carried out. The first sized the magnets and selected the wheel material, because adhesion is the single failure mode that loses the vehicle and its payload. The second addressed steering and braking, because auto-steering on a pole is not trivial.

A. Component Selection

The vehicle is modelled as a rigid body held against a vertical ferromagnetic plane by two magnetic wheels, Fig. 5. Each wheel i experiences a magnetic normal force $f_{m,i}$ directed into the pole, a reaction normal force f_i^n and a tangential friction force f_i^{up} acting against gravity f_g . The model assumes quasi-static ascent, that both wheels remain in contact on a flat of the pole, that the magnetic force is uniform across the contact patch, and that the interface is Coulombic. Table III lists the parameter values used.

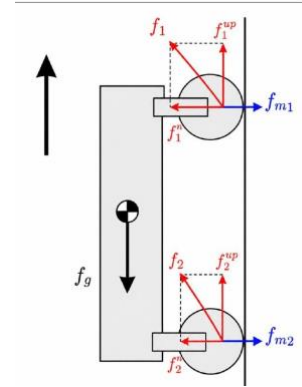


Fig. 5. Free-body diagram of the climbing wheels. The magnetic normal forces f_{m1} and f_{m2} generate reaction normal forces f_1^n and f_2^n , whose associated friction f_1^{up} and f_2^{up} must exceed the weight f_g .

TABLE III. MODELLING ASSUMPTIONS AND FINDINGS

Parameter	Symbol	Value
Target mass	m	1.5 kg
Wheel geometry	n/a	2 wheels, $\text{Ø}50 \times 27.5$ mm
Safety factor	SF	1.5
Coefficient of friction	μ	0.5
Permeability of free space	μ_0	1.2566×10^{-6} H/m
Working flux density	B	1.43 T
Magnet force required	$F_{\text{mag,total}}$	44.15 N

Magnet force produces a normal force, which combines with the coefficient of friction of the wheel material to give friction traction, which in turn produces the vertical climb. The maximum traction force available is set by Coulomb friction acting on the total magnetic normal force,

$$F_{\text{traction,max}} = \mu \cdot F_{\text{mag,total}} \quad (1)$$

and the vehicle climbs only if that traction force exceeds the weight with the chosen safety margin,

$$F_{\text{traction,max}} \geq \text{SF} \cdot mg \quad (2)$$

Combining (1) and (2) gives the adhesion requirement

$$F_{\text{mag,total}} \geq \text{SF} \cdot mg / \mu \quad (3)$$

Substituting $\text{SF} = 1.5$, $m = 1.5 \text{ kg}$ and $g = 9.81 \text{ m/s}^2$ gives a required traction force of 22.07 N, and with $\mu = 0.5$ a required total magnetic pull force of 44.15 N. Divided between the two wheels this is 22.07 N, or 2.249 kgf, per wheel. The magnet face area needed to produce that force follows from the magnetic pressure relation,

$$A = 2\mu_0 F / B^2 \quad (4)$$

which for $F = 22.07 \text{ N}$ and $B = 1.43 \text{ T}$ gives $A = 2.71 \times 10^{-5} \text{ m}^2$, equivalent to a 5.21 by 5.21 mm square of pole face per wheel. This was realised with an N52 neodymium magnet array distributed around each wheel rim [4]. The wheel material was selected from the same relation read backwards. Because the required magnet area scales as the inverse of μ , a compliant rubber tread that holds μ near the rubber-on-steel value of 0.5 is worth more than additional magnet mass.

B. Performance Predictions

Rearranging (1) to (3) for the maximum climbable mass gives $m_{\text{max}} = \mu \cdot F_{\text{mag,total}} / (\text{SF} \cdot g)$, which is linear in both design levers. Fig. 6 plots the two sensitivities. The design point sits at 1.50 kg, three times the 500 g payload requirement. The upper panel shows how brittle that margin is with respect to friction, since a tread that degrades from $\mu = 0.5$ to $\mu = 0.3$ loses two-fifths of the capacity. This prediction drove the decision to re-coat rather than re-magnetise the wheels after the wear failures reported in Section IV.

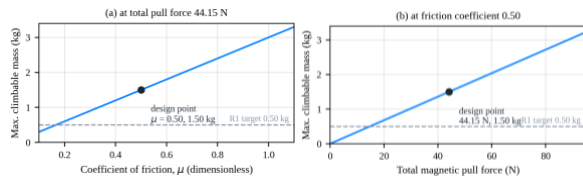


Fig. 6. Predicted maximum climbable mass. (a) Sensitivity to the wheel-to-pole coefficient of friction at a fixed total pull force of 44.15 N. (b) Sensitivity to total magnetic pull force at a fixed μ of 0.5. Both assume a safety factor of 1.5.

Auto-steering raised three further modelling problems: pan-tilt control with a continuous servo, DC motor braking, and the ideal servo brake angle. Panning requires slow motor control so rotating about the pole does not generate excessive centripetal force, and stopping at a chosen position requires precise upward rotation. A 90 degree brake rotation locks firmly but allows a larger positional shift, making the vehicle more prone to falling; a 60 degree brake yaw was therefore selected and implemented as the active-hold brake.

An IMU alone was expected to hold heading, but three failure modes were identified. First, if the robot is upright but not flat on a face of a polygonal pole, the estimator fails to detect the condition, so a suboptimal contact patch is reported as good. Second, it requires calibration before the vehicle is placed onto the pole, which is unacceptable within a 60 s deployment. Third, tilt is measured only relative to that reference, and the magnetometer is greatly affected by the onboard magnets.

The ToF array was therefore promoted to the primary lateral sensor. The 8x8 ranging field is projected onto the pole and the horizontal centre of the pole is estimated as the weighted centroid of the valid cells,

$$x_{\text{pole}} = \sum_i c_i w_i / \sum_i w_i \quad (5)$$

where c_i is the column number of a valid ToF cell, w_i is the weight assigned to that cell and x_{pole} is the estimated horizontal centre of the pole. The rate at which the pole is drifting across the field of view is then

$$R_{\text{raw}} = (x_{\text{pole,k}} - x_{\text{pole,k-1}}) / \Delta t \quad (6)$$

where $x_{\text{pole,k}}$ and $x_{\text{pole,k-1}}$ are the current and previous pole positions, Δt is the time between readings and R_{raw} is the pole movement rate in columns per second. The lateral error $e = x_{\text{pole}} - x_{\text{ref}}$ then drives the steering servos: a negative error steers left, a positive error steers right, and a near-zero error commands straight ahead. Errors are corrected continuously as the robot climbs. Fig. 7 shows the live diagnostic used to tune the loop.

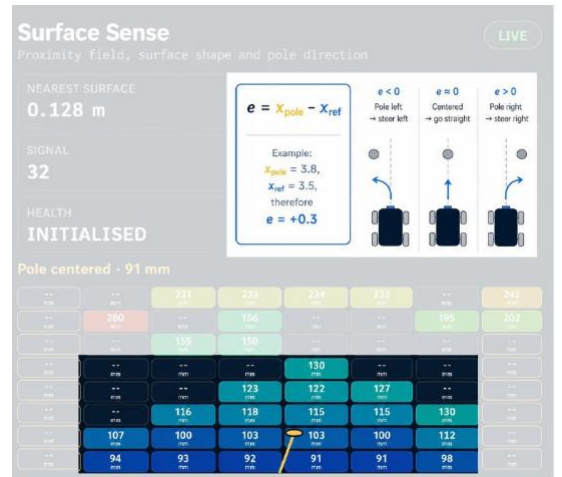
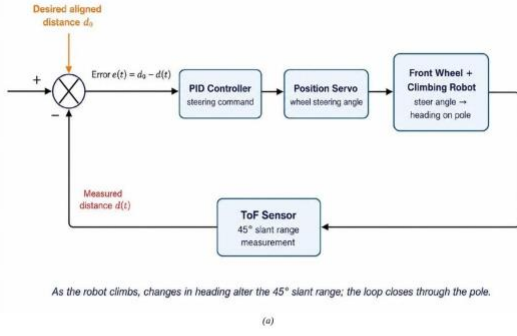


Fig. 7. Time-of-flight surface-sense diagnostic: nearest-surface range, the 8x8 per-cell range field, the estimated pole centre and the resulting steering decision.

The two sensors work in tandem during ascent, Fig. 8. The ToF loop closes first, with a PID controller driving the front steering servo to hold the aligned slant range so the front wheel centres on the pole. Once the ToF residual and pole rate are both low, the IMU rear-wheel loop straightens the chassis behind the centred front wheel, using the stored upright pose as reference. ToF retains authority: if lateral error grows again, the rear IMU correction is cancelled, since lateral position takes priority over chassis straightness.

Closed-Loop ToF Pole Alignment Control

Magnetic climbing robot steering alignment using ToF feedback



IMU Rear-Wheel Feedback with ToF Supervision (Climb Mode)

Rear wheel straightens chassis after ToF has centered the pole

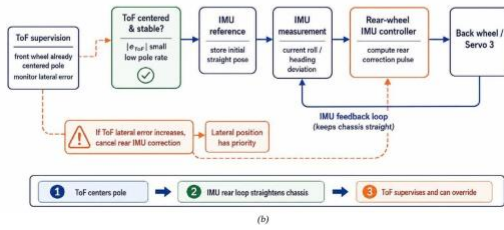


Fig. 8. Control architecture during ascent. (a) Closed-loop ToF pole alignment. (b) IMU rear-wheel feedback under ToF supervision, in which ToF centres the pole, the IMU rear loop straightens the chassis, and ToF supervises and can override.

IV. RESULTS

A. Prototypes

Five major iterations were built, and the team learnt and failed fast across them. V1 and V2 combined an adhesive tread with an early magnetic wheel design. V3 was a four-wheel platform with a battery management system. V4 introduced the ESP32, DRV8833 drivers and a WiFi camera on a dedicated chassis and board, and was then reworked for chassis mass and adhesion with a Halbach wheel and a 2000 mAh lithium-polymer pack before full-payload testing for load, current draw and climb. V5, the version reported here, introduced the MG90 tilt servo, the MG996R steering and the point-tilt-zoom infrared laser. Fig. 9 traces the V5 chassis sequence, from the initial MG90 and MG996 frame through mounted circuit boards and a refined design to full load testing. Corresponding CAD assemblies are shown below each build.

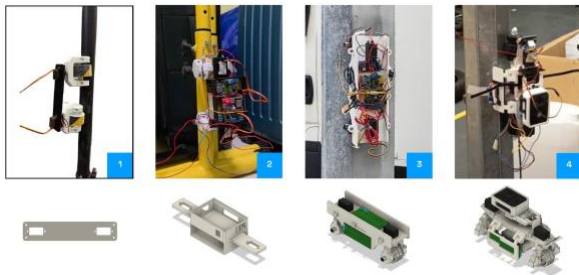
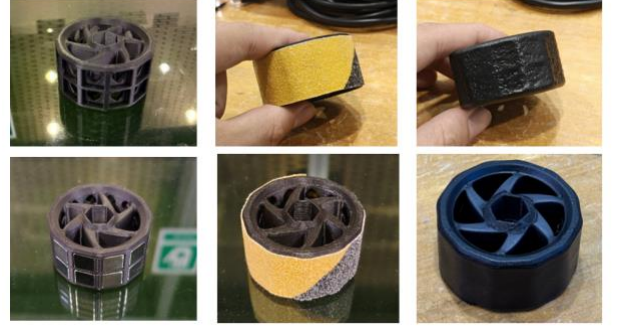


Fig. 9. V5 chassis iterations. (1) MG90 and MG996 frame; (2) mounted circuit boards; (3) refined design; (4) full load testing. Corresponding CAD assemblies are shown below each build.

Two subsystems were iterated independently, Fig. 10. The magnetic wheel combined an N52 array with a spokes design, progressing from a printed rim to a grip-taped variant and finally a sprayed rubberised tread over the tape, retaining its friction on a surface that survives repeated cycles. The point-tilt-zoom subsystem filters the scene into white blobs and

tracks the moving one; this needed the camera's infrared-cut filter replaced with an 850 nm bandpass filter, paired with a 5 W 850 nm infrared laser, after which the pan-tilt loop was PID-tuned.



Point Tilt Zoom



Core Concept:

- Filter into white blobs
- Track moving blob

Lens Modification

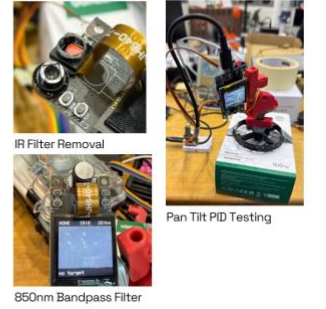


Fig. 10. Subsystem iterations. (a) Magnetic wheel, from the printed N52 array rim through the grip-taped variant to the sprayed rubberised tread. (b) Point-tilt-zoom development: infrared filter removal, 850 nm bandpass filter, designator optics and pan-tilt PID testing.

The electronics for the delivered vehicle carry motor control, a barometer, ToF ranging, an IMU and a power sensor on the VANTAGE v5.2 board set. Enclosure surface modelling and firmware scaffolding were accelerated with a Claude-augmented workflow, in which code implementation and complex surface modelling were outsourced while architecture, feature definition and engineering drawings were retained in-house. The outer chassis reached a printable state in approximately 8 h and the final enclosure in approximately 12 h.

B. Experiments

The objective of the experimental programme was to verify each requirement in Table I against the physical article, and to characterise the failure modes that determine whether the device is safe to leave on a pole.

Test-rig realism was addressed first, Fig. 11(a), using a square pole at the minimum operating diameter (5 cm across a face) and a standard LTA lamppost (7 cm at 1.5 m height), with a safety tether, taped cable management and foam pads to mitigate falling risk during rapid iteration. Reliability was assessed by repeated climb trials, logging success/failure, slip events, climb duration and maximum height. Fig. 11(b) shows the onboard IMU's vertical acceleration through a 4 m ascent, hold and descent: the instrumented run recorded three slip events totalling 1.1 s in a 58 s cycle (mean 0.27 s, peak dip 1.0 m/s²). Across the campaign, slipping duration was 10% with a maximum depth of 4 mm, and the greatest height reached was 5.77 m. Ten ascent/descent cycles over 2 m produced one fall, giving the 90% reliability recorded against R2.

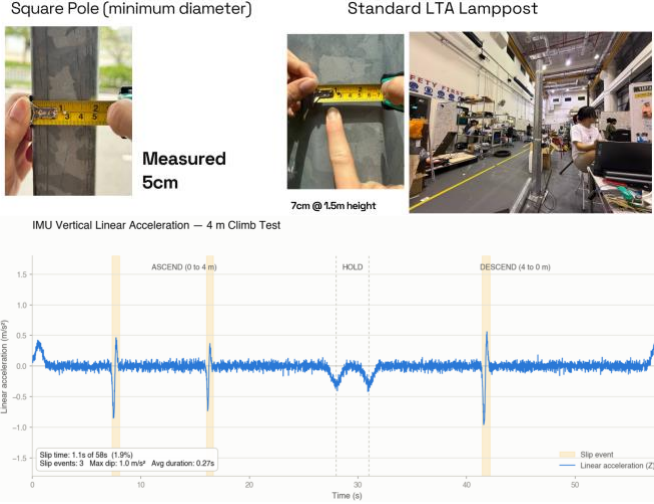


Fig. 11. (a) Testing rig realism: square pole measured at 5 cm across a face and a standard lamppost measured at 7 cm at 1.5 m height, with tether, taped wire and foam pads. (b) Onboard IMU vertical linear acceleration through a 4 m climb test, with detected slip events shaded.

Rapid deployment was measured by timing set-up, timing the climb over a fixed distance, computing the average climbing speed and repeating on both pole profiles. Deployment took 60 s and the vehicle climbed 1 m in 12 s. Sustained climb rates were 0.067 m/s on the square pole, covering 2 m in 30 s, and 0.084 m/s on the octagonal pole, covering 2 m in 24 s. Against R3, a 4 m climb was recorded in 48 s, well inside the 5 min ceiling.

Load bearing was measured in three set-ups on both profiles: powered grip, static hold and dynamic pull. Each result was averaged over three runs. Table IV reports the V5 measured values alongside the V4 prototype's measured values, and Fig. 12 plots them. The worst-case capacity is 1.27 kg against the 0.50 kg required payload, a safety margin of 2.5 to 2.6 times. Averaged across the three set-ups the two profiles are indistinguishable, at 1.51 kg octagonal and 1.50 kg square, so pole shape has negligible effect on capacity. Profile choice is instead driven by the climbing-slip data, where the square profile performs better.

TABLE IV. LOAD-BEARING CAPACITY BY SET-UP AND POLE PROFILE

Set-up	Octagonal V4	Octagonal V5	Square V4	Square V5
Powered grip	123 g	1.61 kg	167 g	1.58 kg
Static hold	10 g	1.27 kg	10 g	1.31 kg
Dynamic pull	500 g	1.65 kg	346 g	1.62 kg

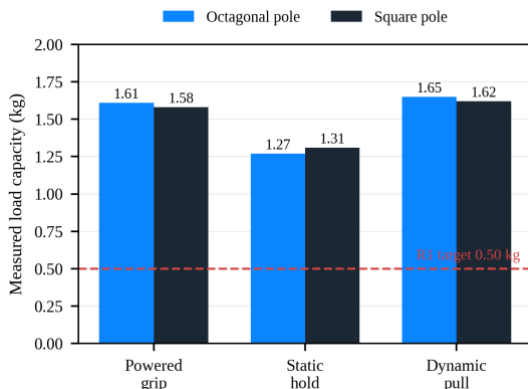


Fig. 12. Measured load capacity by set-up and pole profile, averaged over three runs each, against the 500 g payload requirement.

Table V summarises verification against all seven requirements. Every requirement is met. The two largest margins are payload, at 1.27 kg measured against a 500 g target, and climb time, at 48 s against a 5 min ceiling. Reliability is met exactly at 90%, with no margin.

TABLE V. REQUIREMENT VERIFICATION SUMMARY

#	Parameter	Target	Measured	Status
R1	Payload capacity	≥ 500 g	1.27 kg	Met
R2	Climb reliability	$\geq 90\%$	90%	Met
R3	Time to climb 4 m	≤ 5 min	48 s	Met
R4	Lamppost outer diameter	150 mm	150 mm	Validated
R5	Wireless link range	10 m	10 m	Validated
R6	Battery endurance	≥ 24 h	≥ 24 h	Validated
R7	Pan and tilt cone	$\geq 45^\circ$	$\geq 45^\circ$	Validated

Four failure modes recurred often enough to be designed against, Fig. 13. The servo horn snapped under steering load and was relocated below the frame. Wheel wear from repeated testing was addressed by spraying a thicker rubber coating. Poor wire connections at the motor lead-outs were given a printed stress relief. Insufficient power under simultaneous drive and servo demand was resolved with a higher-output regulator module.

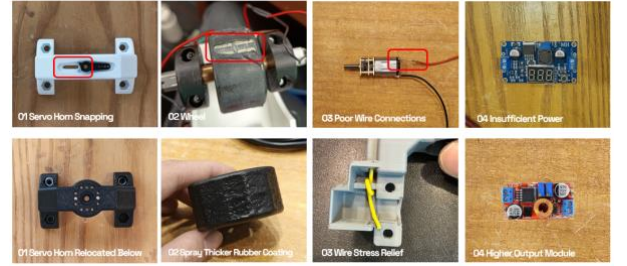


Fig. 13. Failure mode mitigation. Top row: servo horn snapping, wheel wear from testing, poor wire connections and insufficient power. Bottom row: the corresponding mitigations.

V. DISCUSSION

The operational claim behind this project is that elevated overwatch should cost about two minutes and one officer. The measured numbers support it, with 60 s to place and start the vehicle and 48 s to climb 4 m. Compared with the current practice in Fig. 1, where officers strap a camera and hoist to the post by hand, the manpower released is the substantive result rather than the imagery itself. All three SPF must-haves are satisfied.

The most useful single result is that the binding load case is static hold. Because the vehicle holds 1.27 kg with the motors unpowered, a total electrical failure partway up the pole does not drop the payload onto the street below, and the device simply stops. That is a property of permanent-magnet adhesion rather than of the control system, and it is the main reason the magnetic concept was preferred over the aerial alternative. The margin over the 500 g BWC also leaves room for a heavier payload without redesigning the adhesion.

The comparison between the V4 and V5 measured load capacity in Table IV reflects the scale of improvement across iterations rather than a gap in the modelling. V4's measured capacity trailed V5's by an order of magnitude in the powered grip and static hold set-ups, consistent with the adhesion and

chassis rework between the two builds, including the move from V4's Halbach wheel to the N52 magnet-array wheels reported here. The practical consequence is that the final design carries far more margin than the earlier prototype demonstrated, and validating the first-principles model in Section III.B directly against the V5 measurements, rather than against V4, is the first recommendation for follow-on work.

Three limitations qualify these results. First, 90% climb reliability meets R2 exactly and leaves no margin, and the residual is dominated by slip, measured at 10% of climb duration with a maximum depth of 4 mm. Second, testing was conducted on prepared pole sections with a safety tether attached, so weather, surface condition and the ingress protection of the enclosure remain uncharacterised. Third, the concept requires a ferromagnetic pole, so a survey of the intended deployment area is a precondition of use rather than an optional step.

VI. IMPACT AND COLLABORATION

The impact of this work is measured in man-hours rather than camera imagery. Current practice occupies two or more trained officers for a manual emplacement, at the point in an incident when manpower is scarcest. VTG-05 reduces this to one operator who stays on the ground: 60 s to place the device and confirm mounting, and 48 s to climb 4 m. The officers released return to their duties rather than the base of a lamppost, a direct gain in front-line capacity, using equipment and infrastructure the agencies already have.

The requirements for this device were based on real operational requirements. They were obtained through a sequence of engagements with the two front-line agencies that would use it, summarised in Table VI. The pattern across those engagements was consistent: partners were less interested in image quality, which their issued cameras already provide, than in the time and the headcount needed to get a camera to height.

TABLE VI. STAKEHOLDER ENGAGEMENT AND DESIGN OUTCOMES

Date	Engagement	Outcome for the design
4 June	Changi Neighbourhood Police Centre visit	Officers demonstrated the camera systems in service and identified slow deployment and vendor dependence as the core weakness
29 June	Baby Shark Fund award	S\$2,000 towards the build, covering four-fifths of the S\$2,483.18 finally expended
3 July	Follow-up requirements meeting	Consolidated the three must-have parameters: climb 4 m, deploy in under 5 min, carry a 500 g camera
9 July	Army pilot	First demonstration of the vehicle outside the laboratory, on poles the team did not prepare
6 August	Pilot trial	Climb and emplacement exercised end to end in front of the intended users
7 August	Pilot trial and operations feedback	Confirmed the issued body-worn camera as the payload and the single-operator concept of use

Three consequences of that engagement are visible in the design. First, the must-have parameters in Table I are SPF's operational figures rather than team-chosen targets. Second, the mechanical envelope was fixed by field measurement rather than a catalogue: 150 mm nominal outer diameter on the octagonal lamppost, 70 mm across a flat at 1.5 m on a standard LTA post, and 50 mm across a face on the smallest square pole in the set, so testing covered both profiles. Third, the payload interface was fixed as the issued BWC in a printed cradle rather

than a bespoke module, so the robot only has to carry and point a camera officers already use.



Fig. 14. Project team with VTG-05 at a partner demonstration.

That last decision also answers a governance question, made for accountability as much as logistics. A device that can place a camera anywhere within minutes invites the objection that it creates a new, unregulated video source; because the payload is the camera an officer is already issued, the footage lands in the evidence chain that already exists, under the same retention and access rules. The remaining consideration is operational rather than legal: the concept assumes a ferromagnetic pole, and a short survey of the deployment area, a step agencies already take in planning an emplacement, confirms suitability.



Fig. 15. Project team with VTG-05 at the Singapore Police Force pilot trial.

The work has continued beyond the course, supported by a Baby Shark Fund award of S\$2,000 on 29 June against a total spend of S\$2,483.18. The platform has since been taken forward with two industry partners across three demonstration days, the Army pilot in July and police pilot trials in August, both expressing continued interest with a clear path to adoption via the Section V refinements. Crucially, both engagements confirmed the project's founding premise: the constraint on temporary elevated surveillance is the labour of emplacement, not the sensor.

VII. CONCLUSIONS

VTG-05 demonstrates that permanent-magnet wheel adhesion, combined with closed-loop ToF steering, places an issued police body-worn camera at height on existing street infrastructure, quickly and by one operator. A first-principles adhesion model sized the magnets at 44.15 N total pull force for a 1.5 kg vehicle at $\mu = 0.5$ and a 1.5 safety factor. The two-wheel prototype, each wheel driven by a pair of N20 motors

and steered by its own MG996R servo, carries up to 1.27 kg, climbs 4 m in 48 s and reached 5.77 m, meeting all seven engineering requirements within the S\$2,500 budget.

The wider significance is that the expensive part of temporary surveillance is neither the camera, which agencies already issue, nor the pole, which is common street infrastructure they can already use, but the labour of connecting them. Automating that step converts a multi-officer task into a single-operator one and makes elevated overwatch cheap enough to use routinely.

Priority extensions are: reconciling the adhesion model against the measured capacity; raising climb reliability above the 90% floor by improving handling of flat-to-flat transitions; environmental qualification of the enclosure; and a descent-and-recovery mode.

APPENDIX

The project ran over thirteen weeks, Fig. 16: five build iterations compressed into the first twelve weeks, with sensor testing and PCB work running in parallel from week three so failures in adhesion and steering surfaced early. A failure mode and effects analysis and risk register covering fall, slip and power-loss modes ran alongside from week five.

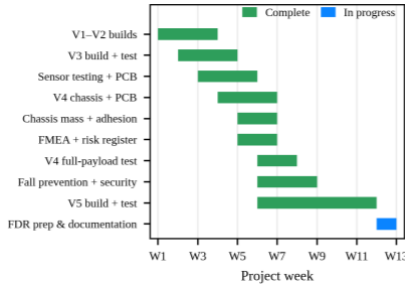


Fig. 16. Iterations timeline, showing the five build iterations and the parallel sensing, failure-mode and documentation streams.

Expenditure totalled S\$2,483.18 against a S\$2,500.00 budget, leaving S\$16.82. Fig. 17 shows the breakdown. Sensing and vision was the largest line at S\$528.04, followed by structure and mechanical at S\$502.65 and actuation and drive at S\$468.91, so the three hardware categories together account for three-fifths of the budget.

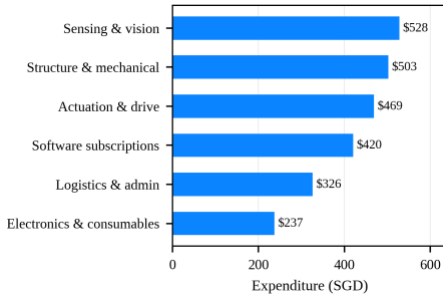


Fig. 17. Expenditure by category. Total S\$2,483.18 against a S\$2,500.00 budget.

TABLE VII. FAILURE-MODE REGISTER

Failure mode	Effect	Cause	Mitigation implemented
Servo horn snapping	Loss of steering authority mid-climb	Horn loaded in bending at the steering interface	Servo horn relocated below the frame

Failure mode	Effect	Cause	Mitigation implemented
Wheel wear from testing	Reduced friction, slip and eventual fall	Abrasion of the tread over repeated cycles	Sprayed thicker rubber coating
Poor wire connections	Intermittent motor drive	Joint fatigue at the motor lead-out	Printed wire stress relief
Insufficient power	Brown-out under simultaneous drive and servo load	Regulator output below peak demand	Higher-output regulator module
IMU heading corruption	False alignment estimate	Magnetometer greatly affected by the onboard magnets	ToF promoted to primary lateral sensor
Fall during testing	Loss of vehicle and payload	Slip on the pole during climb	Safety string, foam pads and 60 degree brake yaw

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